Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/29/2021
PI Name:	Harden, James L. Ph.D.		
Project Title:	A Modular Library of Self-Assembling Artificial Proteins for Three-dimensional Tissue Culture		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1345		
Performance Goal No.:			
Performance Goal Text:			

The focus of this research is on the design and development of a library of artificial proteins for tissue culture applications that capture selected biofunctional aspects of specific extracellular matrix (ECM) proteins. These proteins employ a modular, multi-block architecture that includes independent inter-chain binding, soluble spacer, and bioactive domains, and are engineered to self-assemble from solution into multi-functional, reversible hydrogels. The bioactive domains are chosen from a set of known ECM polypeptide ligands and from new ligands obtained using combinatorial search methods. We are pursuing a rational molecular engineering approach that includes modeling and computer simulation studies to evaluate the molecular properties (conformation, flexibility, and intermolecular binding) of the modular constituents of potential protein designs, systematic experimental characterization of the molecular properties of these synthetic proteins and the materials properties of the hydrogels they form, and studies of cell adhesion and interaction with these hydrogels in two- and three-dimensional culture conditions.

Task Description:	The overall objective of the proposed work is to use de novo protein design strategies, concepts from cell surface receptor biology, and genetic engineering methods to create a library of self-assembling artificial proteins for tissue culture applications. This objective will be addressed by pursuing three interdependent specific aims: 1. Design, synthesis, and characterization of a combinatorial library of self-assembling biofunctional proteins for cell culture substrates, 2. Fabrication and characterization of self-assembled biofunctional hydrogels with controlled composition, microstructure, and materials properties, and 3. Characterization of cellular response to the biofunctional hydrogels and their utility for 3D cell culture. This project should lead to an important new class of biomaterials for three-dimensional tissue culture applications. These gels may be used alone as cell encapsulation materials, or as biofunctional coatings for biodegradable tissue engineering scaffold materials. The ability to control the specificity of cell-matrix adhesion and biochemical signal transduction allow artificial ECM components to be tailored for particular cell types. Moreover, our modular approach also facilitates the simultaneous self-assembly of proteins with different cell-binding functionalities. This provides a route to fabrication of "on-demand " bioactive hydrogels, in which multiple biofunctional properties are obtained through self-assembly of appropriate constituents from a modular protein library. In this manner, matrix materials may be rapidly optimized for specific tissue culture requirements, potentially having a profound effect on tissue engineering methods.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project should lead to an important new class of biomaterials for three-dimensional tissue culture applications. These gels may be used alone as cell encapsulation materials, or as biofunctional coatings for biodegradable tissue engineering scaffold materials. The ability to control the specificity of cell-matrix adhesion and biochemical signal transduction allow artificial ECM components to be tailored for particular cell types. Our modular approach also facilitates the simultaneous self-assembly of proteins with different cell-binding functionalities. This provides a route to fabrication of on-demand bioactive hydrogels, in which multiple biofunctional properties are obtained through self-assembly of appropriate constituents from a modular protein library. In this manner, matrix materials may be rapidly optimized for specific tissue culture requirements, potentially having important effects on tissue engineering methods used in earth-based medicine.
Task Progress:	See publications in Bibliography section.
Bibliography Type:	Description: (Last Updated: 03/29/2021)
Articles in Peer-reviewed Journals	Mi L, Fischer S, Chung B, Sundelacruz S, Harden JL. "Self-assembling protein hydrogels with modular integrin binding domains." Biomacromolecules. 2006 Jan;7(1):38-47. https://doi.org/10.1021/bm050157p ; PMID: 16398496 , Jan-2006
Articles in Peer-reviewed Journals	Aggarwal S, Janssen S, Wadkins RM, Harden JL, Denmeade SR. "A combinatorial approach to the selective capture of circulating malignant epithelial cells by peptide ligands." Biomaterials. 2005 Oct;26(30):6077-86. https://doi.org/10.1016/j.biomaterials.2005.03.040 ; PMID: 15907998 , Oct-2005